# Single-Qubit State Theory & Probability Amplitudes

Comprehensive Theoretical Guide • Derived from Video Analysis (90za6mazNps)

## 1. Definition of a Quantum State

In quantum mechanics and quantum information theory, a **quantum state** provides a complete mathematical description of the physical state or condition of a quantum system. It contains all observable information about how the system is currently configured before any measurement takes place.

## 2. Mathematical Formulation & Computational Basis States

A general single-qubit state vector $|\psi\rangle$ is expressed as a linear combination of its computational basis states:

**General Single-Qubit State Expression:**

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- **Valid Computational Basis States:** For a single qubit, the only valid standard computational basis states are $|0\rangle$ and $|1\rangle$. Other integer state labels (such as $|2\rangle$, $|3\rangle$, or $|10\rangle$) do not represent single-qubit computational basis states.
- **Role of Basis Vectors:** The vectors $|0\rangle$ and $|1\rangle$ form the orthonormal basis spanning the two-dimensional Hilbert space of a single qubit system.

## 3. Probability Amplitudes and the Normalization Rule

In the state expression $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the coefficients **α** and **β** represent the **probability amplitudes** of the quantum state across the respective basis vectors:

**DIAGRAM: MAPPING PROBABILITY AMPLITUDES TO MEASUREMENT OUTCOMES**

| Basis State $|0\rangle$ | Basis State $|1\rangle$ |
|---|---|
| Probability Amplitude: $\alpha$ | Probability Amplitude: $\beta$ |
| Measurement Probability: $P(0) = |\alpha|^2$ | Measurement Probability: $P(1) = |\beta|^2$ |

**Normalization Axiom:** $|\alpha|^2 + |\beta|^2 = 1$

**The Total Probability Rule:** Because measuring a single qubit must always yield either outcome 0 or outcome 1, the physical probabilities $|\alpha|^2$ and $|\beta|^2$ must strictly sum to **1** (representing 100% total probability).

## 4. Core Theoretical Summary

| Quantum Parameter | Mathematical Symbol | Physical Meaning / Rule |
|---|---|---|
| **Quantum State** | $|\psi\rangle$ | Describes the physical state or condition of the system [cite: 19] |
| **Basis States** | $|0\rangle$ and $|1\rangle$ | The two valid single-qubit computational basis states [cite: 19] |
| **Amplitudes** | $\alpha$ and $\beta$ | Complex numbers representing probability amplitudes [cite: 19] |
| **Normalization** | $|\alpha|^2 + |\beta|^2 = 1$ | The total probability of measuring 0 and 1 must equal 1 [cite: 19] |